ANA DIMITRIEVSKA

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Education

University of Waterloo | BASc Mechatronics Engineering **2023 - 2028**

Technical Skills

Controls: Motor Control and Sensor Integration, Raspberry/Arduino, ROS2, MoveIt2, Gazebo/Foxglove, SLAM
Software: Python, OpenCV, Docker, MediaPipe, YOLO, C++/C, Git, SQL
Mechanical: SolidWorks, AutoCAD, Precision Machining, 3D Printing, FEA/ANSYS Analysis, GD&T

Work Experience

Software Development Intern | Red Rabbit Robotics | Vancouver, BC 09/2026 - 12/2026

- Developed and simulated collision-aware T-shirt pick-and-place trajectories using ROS 2 and Gazebo.
- Assembled, wired, and integrated a 6-DOF RealMan robotic arm with in-house software using its Python API.
- Troubleshot sim-to-real integration issues, achieving a 6-second pick cycle through IK, gain, and path planning optimization.

Roboticist Intern | Apera-AI | Vancouver, BC 01/2026 - 04/2026

- Programmed 6-DOF Universal Robots and FANUC robots for automated pick-and-place operations using vision pipelines.
- Shipped a Docker deployed Python app to automate camera calibration and focusing workflows using ChArUco board detection.
- Resolved customer-reported issues by updating KAREL robot driver software, improving system reliability and functionality.

Robotics Research Assistant | Waterloo Interactive Robotics Lab 09/2025 - 12/2025

- Developed ROS2 control software for a 7-DOF PAL robot arm, integrating gripper control for custom force sensors.
- Simulated a robotic pick-and-place pipeline in Gazebo using MoveIt2 for trajectory planning and motion execution.
- Designed and 3D printed camera mounts and test hardware to support robot testing, remote operation, and software debugging.

Software Development Intern | Schneider Electric | Montpellier, France 05/2025 - 08/2025

- Leveraged Python to develop a PyQt6-based automation tool that creates Network Diagram Schematics, reducing overhead by 25%.
- Optimized app performance by 30% using multithreading and automated UI testing with Selenium.
- Collaborated with a senior developer to train an AI model using OpenCV and PyTorch on electrical diagram specifications.

Software Development Intern | Schneider Electric | Montpellier, France 09/2024 - 12/2024

- Developed a Python app to automate power systems equipment testing, driving demand and adoption by teams in the U.S. and U.K.
- Created a Python-based test script generation tool used by ~30 engineers to simulate and validate over 500 electronic devices.
- Debugged and integrated a C++ automation tool into Python, avoiding a full rebuild and saving 20+ hours of development work.

Extracurriculars

Robot Hand | Personal Project 02/2026 - 04/2026

- Prototyped a 3D-printed tendon-driven robotic hand with actuated fingers and a multi-axis thumb using 7 motors and an Arduino.
- Developed a Python/MediaPipe vision pipeline to detect and replicate human hand motion, successfully grasping various objects.
- Optimized actuator design and circuit integration to reduce torque demands and increase motor control precision.

Robot Navigation | Watanomous Design Team 09/2025 – 12/2024

- Developed ROS2 autonomous navigation for a simulated differential-drive robot using camera and LiDAR sensor data.
- Implemented waypoint tracking, obstacle avoidance, sensor fusion, filtering, and buffering for reliable navigation
- Tested and debugged SLAM-based navigation workflows in Foxglove and Gazebo.